Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	**Development of an AI-Enabled Semantic Edge for Vendor-Heterogeneous OT Data Interpretation - A Semantic Interoperability Framework for Agentic AI in Manufacturing Environments**			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (√), Next-gen Communication ( ), Advanced Robotics & Manufacturing (√), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1 (**Large Category**)**		**Technology Field 2 (**Large Category**)**	
	Information & Communication			
	Medium Category	Small Category	Medium Category	Small Category
	Software	Software Solutions		
Concept of Proposed Technology	The proposed technology aims to realize an AI-enabled Semantic Edge that enables artificial intelligence systems or Agentic AI to directly understand and reason over operational technology (OT) data generated in manufacturing environments, with a particular focus on lowering the barrier to AI adoption for Korean small and medium-sized enterprises (SMEs). In current manufacturing sites, especially those operated by SMEs, OT data is produced by a wide variety of equipment supplied by different vendors, each relying on proprietary protocols and data models. Although such data is syntactically accessible, its meaning remains largely implicit and vendor-dependent, making it difficult for AI systems to be deployed, reused, or scaled without significant custom engineering effort. **Core Concept** The core concept of this technology is to introduce a semantic layer at the edge, positioned between			

	protocol adapters and AI services, which explicitly represents the meaning of OT data in a machine-interpretable form. This semantic layer performs two essential functions: 1. Semantic lifting, which transforms vendor-specific OT data into explicit semantic representations that describe what the data represents in terms of assets, states, capabilities, units, constraints, et cetera,; and 2. Context-aware semantic interpretation, which dynamically interprets these representations based on current operational conditions such as process phase, operating mode, configuration, and temporal context. In particular, the semantic lifting process is designed around a digital asset-centric information structure, enabling OT data to be organized, described, and accessed in a consistent and extensible manner across heterogeneous equipment. This approach allows SMEs to abstract vendor-locked OT data without requiring intrusive changes to existing shop-floor systems, while remaining aligned with emerging international standardization directions in industrial digitalization. For Korean SMEs, the proposed AI-enabled Semantic Edge fundamentally lowers the barrier to industrial AI adoption by transforming vendor-locked OT data into AI-ready semantic representations. This enables SMEs to deploy, reuse, and scale AI solutions without repeatedly redesigning data pipelines for each piece of equipment or production line. As a result, AI transitions from isolated proof-of-concept deployments to a reusable, scalable, and economically viable capability for SMEs. By embedding these semantic capabilities directly at the edge, AI systems no longer consume raw OT data, but instead operate on contextually meaningful, vendor-agnostic information. This architectural shift decouples AI services from proprietary OT data models and provides a practical foundation for industry-oriented AI services, agent-based applications, and future smart manufacturing solutions.
Rationale for Joint R&D Project	**Critical Technical Barriers** Despite growing demand for AI-driven smart manufacturing, industrial AI adoption remains largely limited to isolated proof-of-concept projects. A key reason is that semantic interoperability in OT environments cannot be achieved through engineering integration alone. On the industrial side, equipment vendors, edge device suppliers, and AI service providers each operate within their own technical boundaries, often deferring responsibility for semantic modeling to one another. While standardization is frequently requested, existing standards are inherently “ex post” and do not fully address the level of semantic rigor required for machine reasoning and autonomous AI operation. Semantic interoperability in OT domains is fundamentally a research-intensive problem, requiring the integration of formal semantics, knowledge representation, reasoning mechanisms, and cyber-physical systems engineering (so-called Digital Twin), all under strict real-time and safety constraints. This joint R&D project is therefore necessary to combine: • the academic research capabilities of an international research institute with expertise in CPS, industrial systems, smart manufacturing and AI, and • the practical industrial environments and deployment need as well as productization capability of Korean SMEs.

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	Through this collaboration, the project aims to establish a principled and practically validated foundation for OT-first semantic interoperability, enabling AI-enabled Edge systems that can be broadly adopted by SMEs. **Why Global Collaboration?** Purdue University, a leading overseas research institution, is a manufacturing-focused engineering university located in Indiana, a region surrounded by numerous manufacturing companies, and serves as an industrial hub in the United States. The university has already been collecting meaningful equipment signal data and OT data from major manufacturing corporations in the surrounding area and utilizing them for AI-based research. (1) Purdue maintains strong collaborative relationships with medium- to large-scale enterprises equipped with factories, facilities, and production lines; (2) it operates specialized research centers capable of conducting testbed-based research on manufacturing and physical AI; and (3) it is strategically located in a major transportation hub in the U.S., making it the only academic institution where Korean companies can directly identify technological trends and market demands in the local environment. Therefore, global collaboration is essential. In addition, Purdue possesses core technologies related to heterogeneous data pipelines and industrial protocols such as MTConnect, OPC-UA, PROFINET, and EtherCAT, and has extensive expertise and insights in applying AI and LLM technologies. As a result, strong synergies with Korean SMEs are expected.
Key Words	1. Semantic Interoperability 2. Semantic Lifting 3. AI-Enabled Edge 4. OT Data Interpretation 5. Asset-Centric Information Modeling

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (4)	End of the Project	Level (6)

R&D Objectives

R&D Objectives

The primary objective of this joint R&D project is to develop a semantic interoperability framework at the edge that enables artificial intelligence systems to directly understand, interpret, and reuse OT data generated in vendor-heterogeneous manufacturing environments.

Detailed Objectives

More specifically, the project aims to achieve the following objectives:

1. To develop semantic lifting technologies that transform vendor-specific, protocol-dependent OT data into explicit, machine-interpretable semantic representations suitable for AI processing.
2. To establish context-aware semantic interpretation mechanisms that dynamically interpret OT data semantics based on operational context at runtime.
3. To integrate these semantic capabilities into an AI-enabled Semantic Edge prototype that can be deployed in SME-scale manufacturing environments.
4. To validate the effectiveness of the proposed approach through representative industrial use cases relevant to Korean SMEs, demonstrating improved AI reusability, scalability, and deployment efficiency.

The project will target the following technical performance indicators:

- Support for multiple heterogeneous OT data sources originating from equipment supplied by different vendors.
- Successful semantic lifting of OT data into asset-centric, vendor-agnostic semantic representations.
- Context-aware semantic interpretation accuracy under changing operational conditions (e.g., process mode changes, equipment state transitions).
- Edge-level processing latency that meets real-time or near-real-time operational requirements.
- Demonstrated ability to reuse AI models across different equipment or production lines without redesigning data pipelines.

R&D Target Specifications

The target specifications of this project are:

- Vendor support: more than 3 heterogeneous OT data sources originating from equipment supplied by different vendors
- Industrial protocol support: more than 5 different industrial protocols
- Processing latency: end-to-end semantic processing latency below 100 milliseconds in the edge device
- Prototype development: 1 standalone edge device

	• AI reuse: same model across more than 2 scenarios
R&D Plan	The R&D plan is structured around two tightly coupled research tracks and one implementation track that together form the semantic core of the AI-enabled Edge. Task 1(research track): Semantic Lifting of Vendor-Heterogeneous OT Data Objective: To develop a semantic lifting framework that transforms vendor-specific OT data into explicit semantic representations that can be directly consumed by AI systems Key Activities: • Analysis of representative OT data models and protocols commonly used in SME manufacturing environments • Design of a foundational, asset-centric semantic model for describing industrial equipment, states, events, capabilities, and constraints • Development of semantic lifting mechanisms that map vendor-specific OT data to semantic representations using rule-based and hybrid approaches • Implementation of lightweight semantic data structures optimized for deployment at the edge Expected Outputs: • Asset-centric semantic model for OT data abstraction • Semantic lifting rules and mapping mechanisms for heterogeneous OT data • Laboratory-scale validation of semantic lifting accuracy and consistency Task 2(research track): Context-Aware Semantic Interpretation for Edge AI Objective: To enable dynamic interpretation of OT data semantics based on operational context, ensuring reliable and situation-aware AI inference at the edge Key Activities: • Identification and modeling of key OT context dimensions, including process phase, operating mode, configuration, temporal conditions, and maintenance state • Design of a context ontology and lightweight reasoning mechanisms suitable for edge environments • Development of runtime context-switching mechanisms that adjust semantic interpretation based on changing operational conditions • Integration of context-aware semantic interpretation with AI input pipelines Expected Outputs: • Formal context ontology for OT environments • Prototype context-aware semantic interpretation engine deployable at the edge

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	• Experimental evaluation of semantic interpretation robustness under dynamic contexts Task 3(implementation track): Integration and Prototype Validation of AI-enabled Semantic Edge Objective: To integrate semantic lifting and context-aware interpretation into a unified AI-enabled Semantic Edge prototype and validate its applicability in SME-oriented scenarios Key Activities: • Integration of Task 1 and Task 2 outcomes into a unified edge architecture • Development of interfaces between the Semantic Edge and AI services or AI agents • Validation using representative SME manufacturing scenarios, such as condition monitoring, anomaly detection, or process optimization • Evaluation of AI deployment efficiency, reusability, and scalability compared to conventional edge architectures Expected Outputs: • Prototype AI-enabled Semantic Edge system • Demonstration of AI model reuse across heterogeneous equipment • Guidelines for deploying semantic edge technology in SME manufacturing environments

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	The international research institute (Purdue University) will serve as the academic and methodological partner, providing the theoretical foundations and advanced research contributions required to establish OT-first semantic interoperability at the edge. The expected roles and responsibilities of the research institute include: • Contributing to the conceptual design and theoretical formulation of asset-centric semantic models for OT data abstraction • Jointly researching semantic lifting methodologies, including ontology design principles and mapping strategies for vendor-heterogeneous OT data • • Research on data parsing technologies and intelligent decision-making models based on generative AI and multi-agent LLM frameworks • Establishment of graph network–based ontology design strategies • Developing and validating context-aware semantic interpretation mechanisms, including formal context modeling and reasoning approaches suitable for industrial edge environments • Providing academic insight into semantic consistency, reasoning correctness, and system-level implications of semantic interoperability in cyber-physical systems • Supporting laboratory-scale validation, evaluation methodologies, and documentation of research outcomes

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	• Collaborating with the industrial partner to ensure alignment between academic research results and practical implementation constraints • Through these activities, the research institute will ensure that the project outcomes are grounded in rigorous academic research and aligned with international research and standardization trends.
Expected R&R of Korean SME Partner	The industrial partner (Korean SME) will take primary responsibility for the practical implementation, system integration, and productization of the AI-enabled Semantic Edge. The expected roles and responsibilities of the industrial partner include: • Leading the design and implementation of the AI-enabled Semantic Edge device, including hardware architecture, embedded software, and system integration • Implementing semantic lifting and context-aware semantic interpretation functions as executable components within the edge device, based on jointly developed research outcomes • Integrating protocol adapters, semantic layers, and AI/agent interfaces into a deployable and maintainable edge product • Conducting pilot deployments and performance validation in representative SME manufacturing environments • Refining system architecture and functionality based on operational feedback, with a clear focus on reliability, scalability, and cost-effectiveness for SMEs • Driving productization and commercialization, including preparation for manufacturing, packaging, documentation, and market introduction of the AI-enabled Semantic Edge device As the project executing entity, the industrial partner will ensure that the research results are translated into a tangible, market-ready product that can be adopted by Korean SMEs.

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	**Technical Outcomes:** The proposed joint R&D project is expected to deliver the following key technical outcomes: • Development of an AI-enabled Semantic Edge device that embeds semantic lifting and context-aware semantic interpretation capabilities for OT data • Establishment of an asset-centric, vendor-agnostic semantic abstraction layer that enables AI systems to directly consume and reason over OT data • A deployable edge architecture that decouples AI services from proprietary OT protocols and data models • Validation of semantic interoperability techniques under realistic industrial constraints, including real-time operation and heterogeneous equipment environments These outcomes will provide a concrete technical foundation for scalable and reusable industrial AI solutions. **Commercialization and Product Impact:** As the project executing company, the industrial partner will translate the research results into a commercially viable AI-enabled Semantic Edge product. The expected commercialization impacts include: • A market-ready edge device that can be integrated into existing SME manufacturing environments with minimal disruption • Reduction of deployment complexity and cost for industrial AI solutions by providing AI-ready semantic interfaces at the edge • Creation of a modular product platform that can support multiple AI applications, such as condition monitoring, anomaly detection, and process optimization By productizing semantic interoperability as an edge device, the project enables a clear pathway from research outcomes to market adoption. **Impact on Korean SMEs:** Although Korean SMEs do not directly participate as end users in this project, they represent the primary beneficiaries and target market of the proposed technology. The expected impacts on SMEs include: • Lower barriers to AI adoption, as SMEs can deploy AI solutions without redesigning OT data pipelines for each piece of equipment or production line • Increased reusability and scalability of AI models, enabling SMEs to expand AI usage beyond isolated proof-of-concept projects • Reduced dependency on vendor-specific systems, supporting greater flexibility in equipment selection and factory upgrades • Access to AI-enabled manufacturing capabilities through a productized, cost-effective edge solution tailored to SME constraints These impacts directly align with the policy objectives of the Ministry of SMEs and Startups by

	enabling practical and sustainable digital transformation for SMEs. Broader Industrial and Economic Impact: Beyond individual SMEs, the project is expected to generate broader industrial and economic benefits such as: • Acceleration of AI-driven manufacturing innovation by establishing semantic interoperability as a reusable infrastructure layer • Strengthening the domestic industrial AI ecosystem by enabling AI solution providers to develop vendor-agnostic, scalable products • Contribution to international competitiveness by aligning with global research and standardization trends in industrial digitalization The project thus supports not only technological advancement but also long-term industrial competitiveness and ecosystem development.

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6. References

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